

AAKASH SANGANI

AI governance and auditable AI research

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RESEARCH INTERESTS

Auditable and accountable AI systems. Current threads: inferring the revealed objectives of deployed AI agents from their behaviour, algorithmic fairness in Indian credit markets, and conformity assessment and incident reporting for AI under the EU AI Act and California's SB-53.

EDUCATION

D. J. Sanghvi College of Engineering, University of Mumbai

2019 – 2023

B.Tech, Information Technology. CGPA 8.90 / 10 (credit-weighted across 8 semesters)

- Merit-based scholarship covering 100% of tuition.
- **Relevant coursework:** Probability & Statistics (O, 93%), Statistical Analysis (A+, 81%), Artificial Intelligence (A+, 84%), Advanced Data Structures (O, 95%), Design and Analysis of Algorithms (A+, 83%), Discrete Structures (A+, 84%).

RESEARCH & INDEPENDENT WORK

Inverse-Objective Regulator (IOR)

github.com/aakash2410/Inverse-Objective-Regulator

Designed and built a black-box AI-safety auditing system; sole author of the accompanying arXiv paper

- Infers a deployed AI agent's revealed objective from its tool-call traces via Bayesian inverse reinforcement learning, scores divergence from its declared purpose, and emits regulator-legible evidence mapped to the EU AI Act, OWASP, MITRE ATLAS, and NIST AI RMF.
- Released an open 8-agent benchmark spanning agentic safety failure modes (reward hacking, power-seeking, sycophancy, privacy leakage, safety-skipping); a deterministic open cross-encoder critic recovers planted objective divergence on 6/8 agents (mean $r = 0.55$), beating a frontier LLM-as-judge (mean $r = 0.08 \rightarrow 0.43$) with near-zero variance and no API cost.
- Sole author of the accompanying arXiv system paper and full Python implementation (Bayesian IRL, LLM/decision-model judges, reproducible gym, JSON/SARIF reporting).

Credit Equity Audit of India (NSSO 77th Round)

github.com/aakash2410/CreditAudit-India

Independent fairness audit of household credit access and allocation

- Two-stage, Heckman-style deep learning framework separating access to the credit market from the allocation of formal versus informal credit among households already in it.
- Female-headed households had about 59% of the probability of entering the credit market compared with male-headed households (disparate impact 0.59).
- Applied IBM AIF360 adversarial debiasing and SHAP to test for proxy discrimination, reaching near-ideal disparate impact at ~73.2% accuracy. Documented the limits of the design: self-reported survey data, collider bias risk, and SHAP as a diagnostic of the model rather than causal evidence.

rag-compliance-auditor

github.com/aakash2410/MCP_RAG_Compliance

Sole-authored open-source package, published on PyPI

- MCP server that certifies RAG and AI pipelines against the EU AI Act, GDPR/DPDP, HIPAA, SEBI/RBI, and NIST AI RMF across 7 audit dimensions.
- Issues RS256-signed JWT certificates, exposes 8 MCP tools, and runs as a CI/CD gate with SARIF reporting.

PUBLICATIONS

Book chapter

2022 – 23

Sangani, A., Doshi, P., & Sawant, V. "Unusual Social Media Behavior Detection Using Distributed Data Stream Mining." In *Practical Data Mining Techniques and Applications*. Auerbach Publications (CRC Press).

- Proposed a clickstream-based system to detect hacked or bot-controlled social media accounts: users clustered by behaviour, with a Very Fast Decision Tree (Hoeffding-based) classifier per cluster, parallelised across Apache Spark.
- Built in concept-drift handling so that a genuine change in a user's behaviour is relabelled and used to retrain the model instead of being flagged as a compromised account.

SELECTED ACADEMIC PROJECTS

Baby Cry Reason Detection

2022 – 2023

Final project (team of 3), D. J. Sanghvi College of Engineering. Guide: Prof. Prachi Tawde

- Audio classifier for the cause of an infant's cry (belly pain, burping, discomfort, hunger, tiredness) on a 457-sample dataset, using temporal, spectral, cepstral, and prosodic features (MFCCs, spectral contrast, pitch).

- Addressed class imbalance with SMOTETomek; final model was a soft-voting SVM + KNN ensemble, deployed in a Flutter app with a feedback loop for retraining.

Personality Prediction with CV and Social Media Analysis

2022

Team project (team of 4), D. J. Sanghvi College of Engineering. Guide: Neha Ram

- Combined CV text analysis with Facebook behavioural data (myPersonality dataset, 250 users, 9,917 status updates) to predict the Big Five traits.
- A 7-layer LSTM reached 0.82 – 0.85 accuracy, against 0.45 – 0.61 for SVM, Logistic Regression, Gradient Boosting, and XGBoost baselines.

PROFESSIONAL EXPERIENCE

Kalpa Impact

Feb 2026 – present

AI Consultant

Mumbai

- Digital public infrastructure and AI governance consulting for UNDP, DPGA, Gates Foundation, and Co-Develop.
- Collaborated with the Principal Scientific Adviser's Office to write a section in "AGI Governance" on AI incident reporting mechanisms India can adopt, based on the EU AI Act and California's SB-53.
- Sole PM and builder on an offline RAG system, in collaboration with the Government of India's Bhashini.

ZS Associates

Sep 2023 – Jan 2026

Business Technology Solutions Associate, promoted to Senior Associate

Pune

- Launched an agentic AI analytics product for natural-language querying of sales-performance metrics. Owned incentive-compensation product logic for a Fortune 100 pharma client at 100% accuracy.

Bright Money

Jul 2022 – Feb 2023

Product Analyst Intern

Bangalore

- Built A/B testing infrastructure and SQL dashboards. Launched a referral programme that saved \$120K in customer-acquisition cost over 5 months.

TECHNICAL SKILLS

Methods: Bayesian inverse reinforcement learning, LLM evaluation and red-teaming, algorithmic fairness (AIF360, SHAP), A/B testing, RAG, MCP servers

Frameworks: EU AI Act, NIST AI RMF, ISO 42001, GDPR/DPDP, HIPAA, OWASP ASI, MITRE ATLAS

Tools: Python, SQL, AWS, PostgreSQL, Docker, pandas, scikit-learn, TensorFlow

HONOURS

- Merit-based scholarship, 100% tuition waiver, D. J. Sanghvi College of Engineering (2019 – 2023).
- Recognised as a top 1% performer firm-wide at ZS Associates; promoted to Senior Associate within 8 to 10 months of joining.